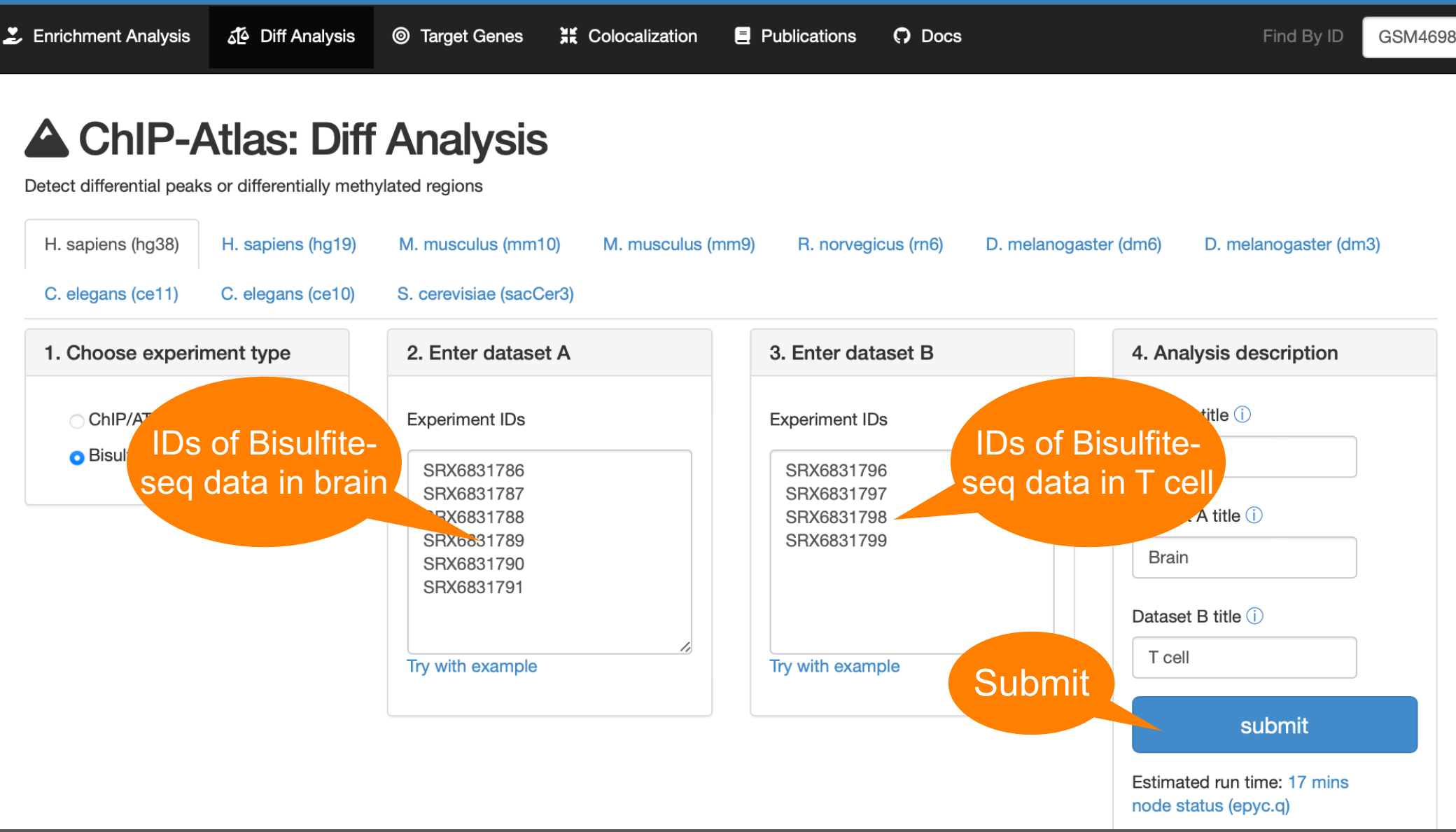
The manual for

ChIP-Atlas: Diff Analysis

1. Overview
2. Differential peak regions
   (ChIP/ATAC/DNase-seq data)
3. Differentially methylated regions
   (Bisulfite-seq data)
4. Tips

Overview of the Diff Analysis tool



The ChIP-Atlas “Diff Analysis” tool can be used to identify differential peak regions or differentially methylated regions from two given sets of ChIP/ATAC/DNase-seq or Bisulfite-seq data, respectively.

Overview of the Diff Analysis tool



The “Diff Analysis” tool can be used to identify differential peak regions or differentially methylated regions from two given sets of ChIP/ATAC/DNase-seq or Bisulfite-seq data, respectively.

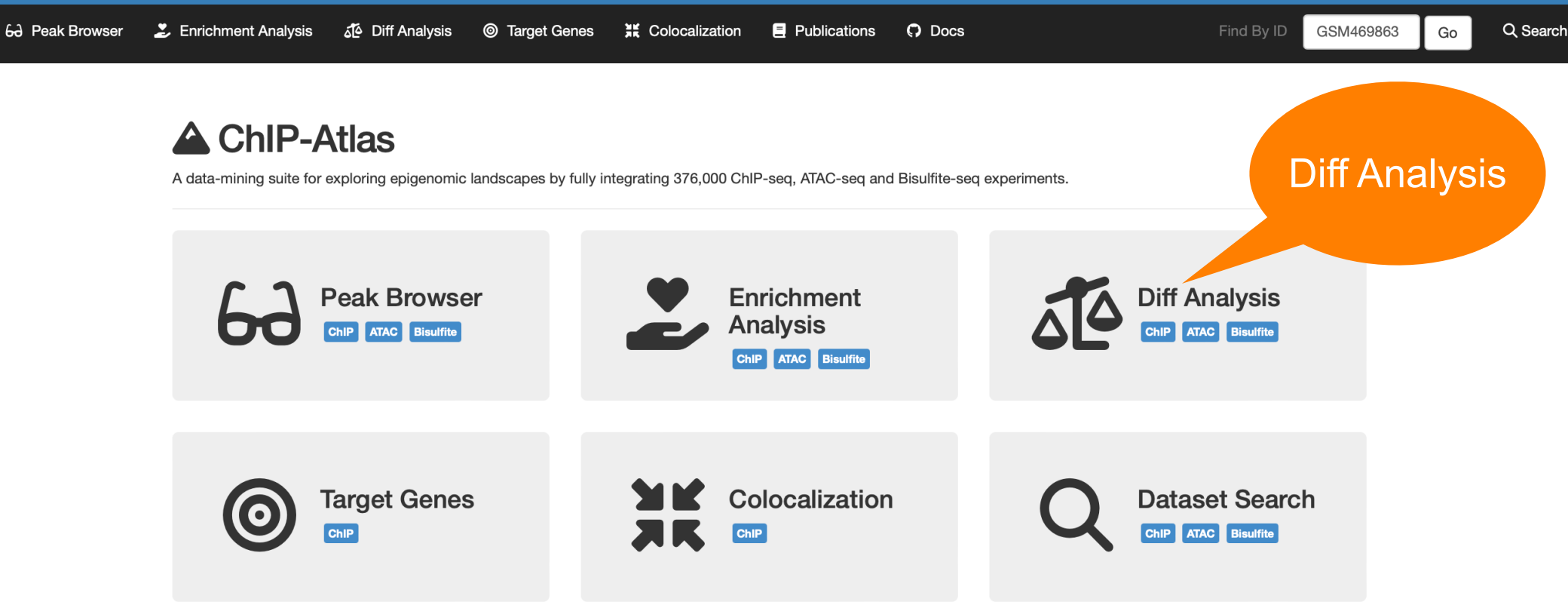
The manual for

ChIP-Atlas: Diff Analysis

1. Overview
2. Differential peak regions
   (ChIP/ATAC/DNase-seq data)
3. Differentially methylated regions
   (Bisulfite-seq data)
4. Tips

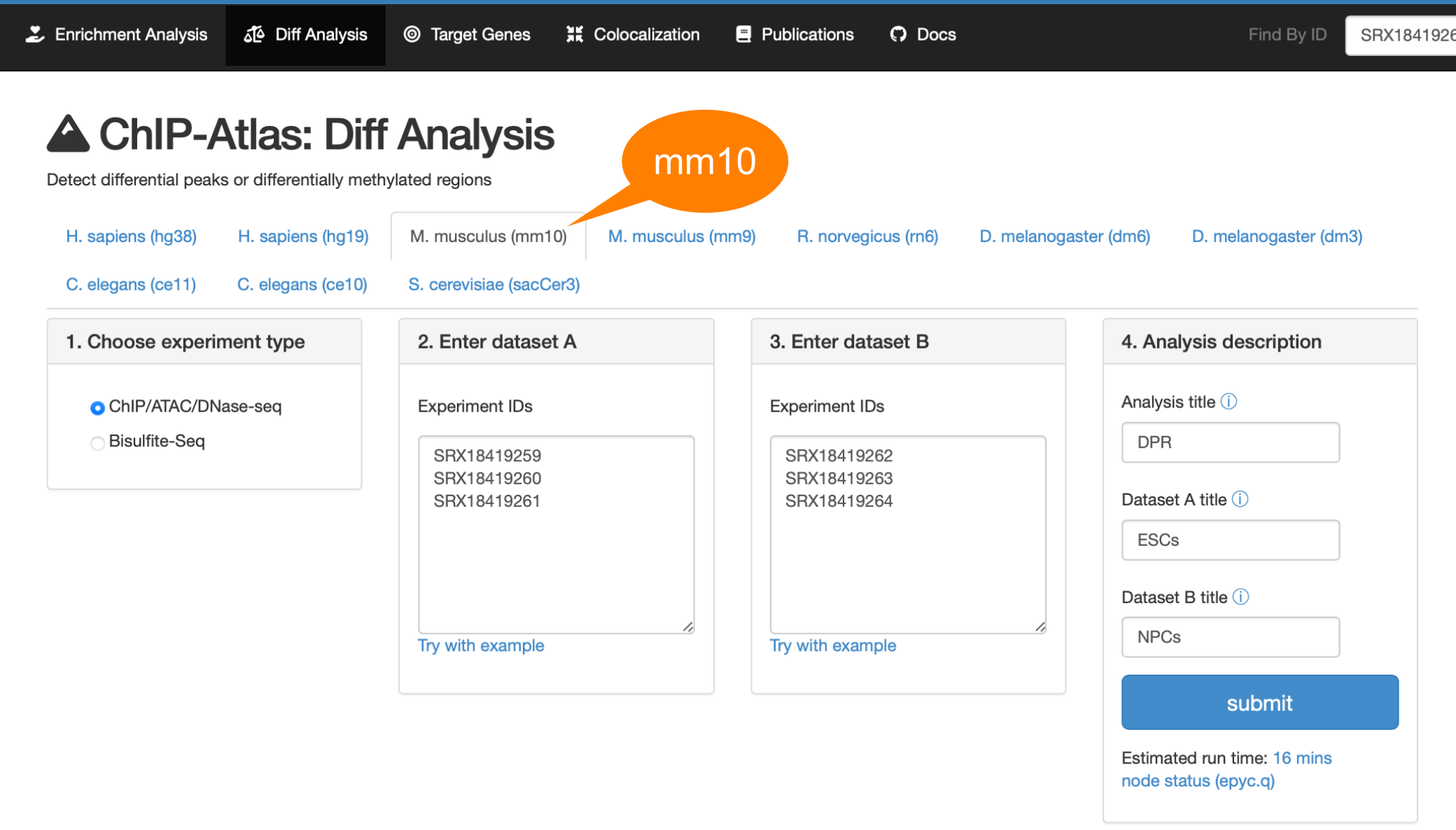
Here, let us try to use the ChIP-Atlas “Diff Analysis” tool to identify differential peak regions (DPRs) from two sets of ATAC-seq data in mouse embryonic stem cells (ESCs) and neural progenitor cells (NPCs).

Diff Analysis tool



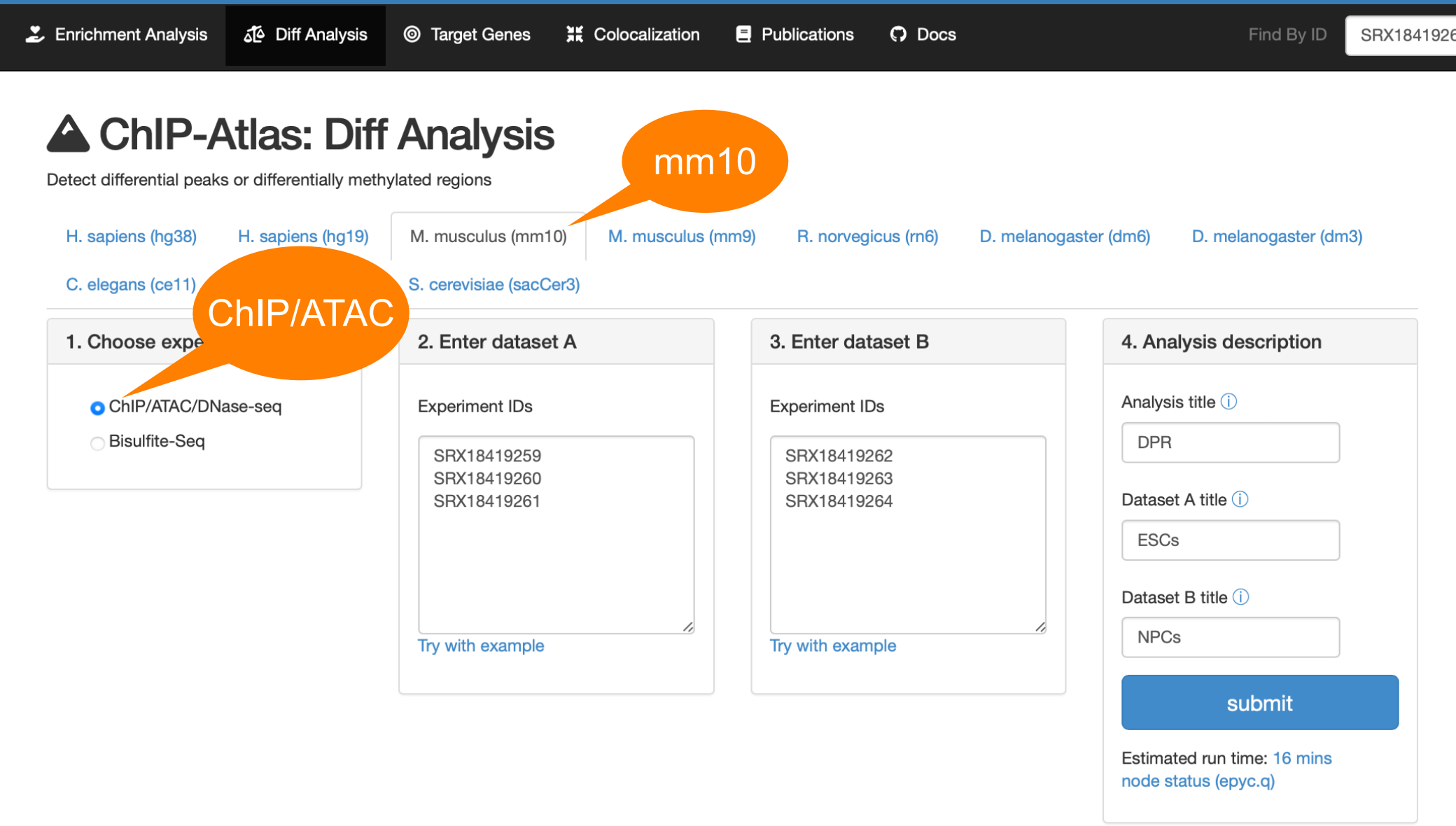
Access the ChIP-Atlas home page (https://chip-atlas.org) and click on the “Diff Analysis” panel.

Settings to identify differential peak regions (DPRs)



Choose a genome assembly of your interest.

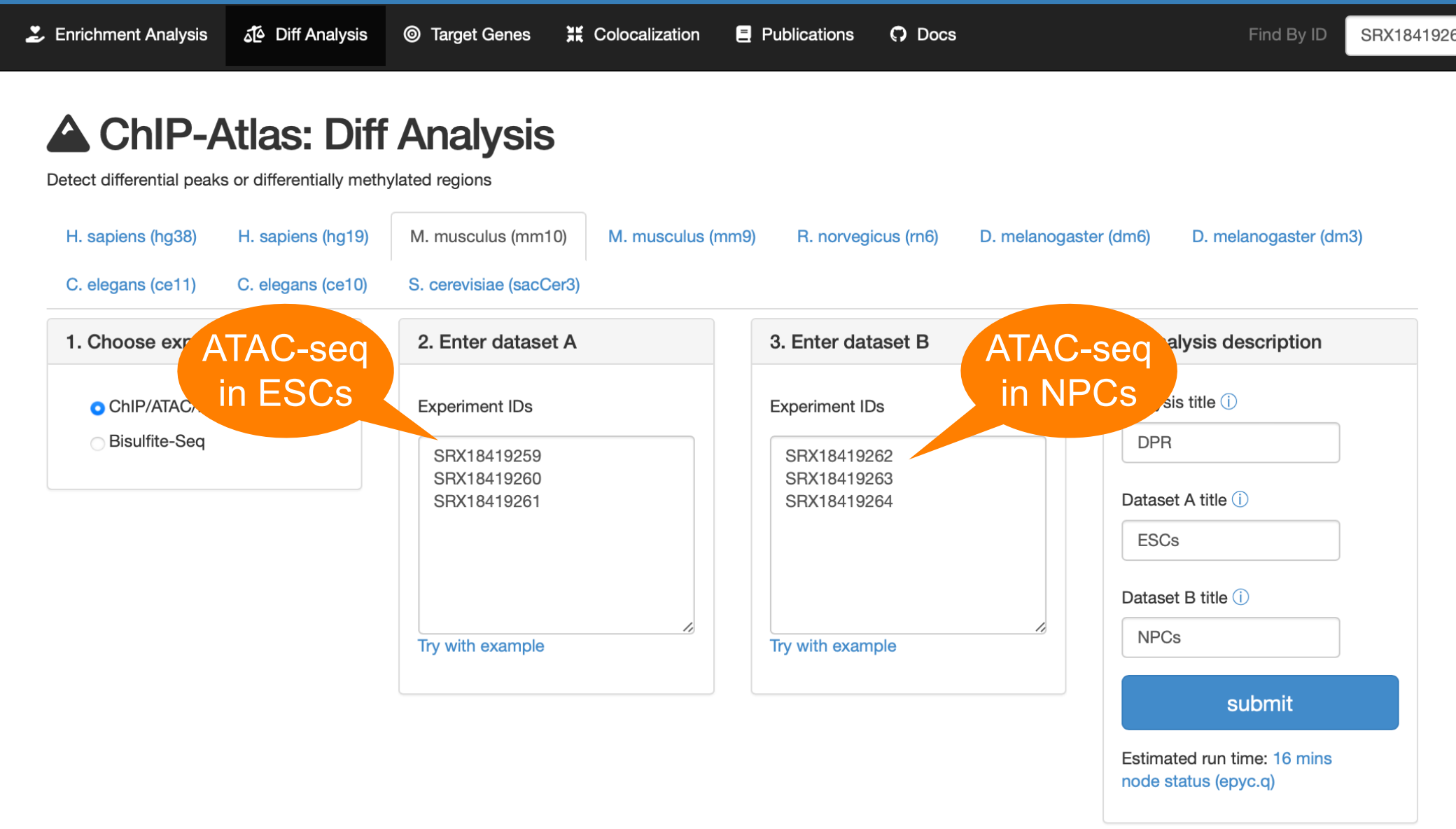
Settings to identify differential peak regions (DPRs)



Check the “ChIP/ATAC/DNase-seq” button to obtain DPRs from two given sets of ATAC-seq, ChIP-seq, or DNase-seq data.

Tips: Methods for analysis

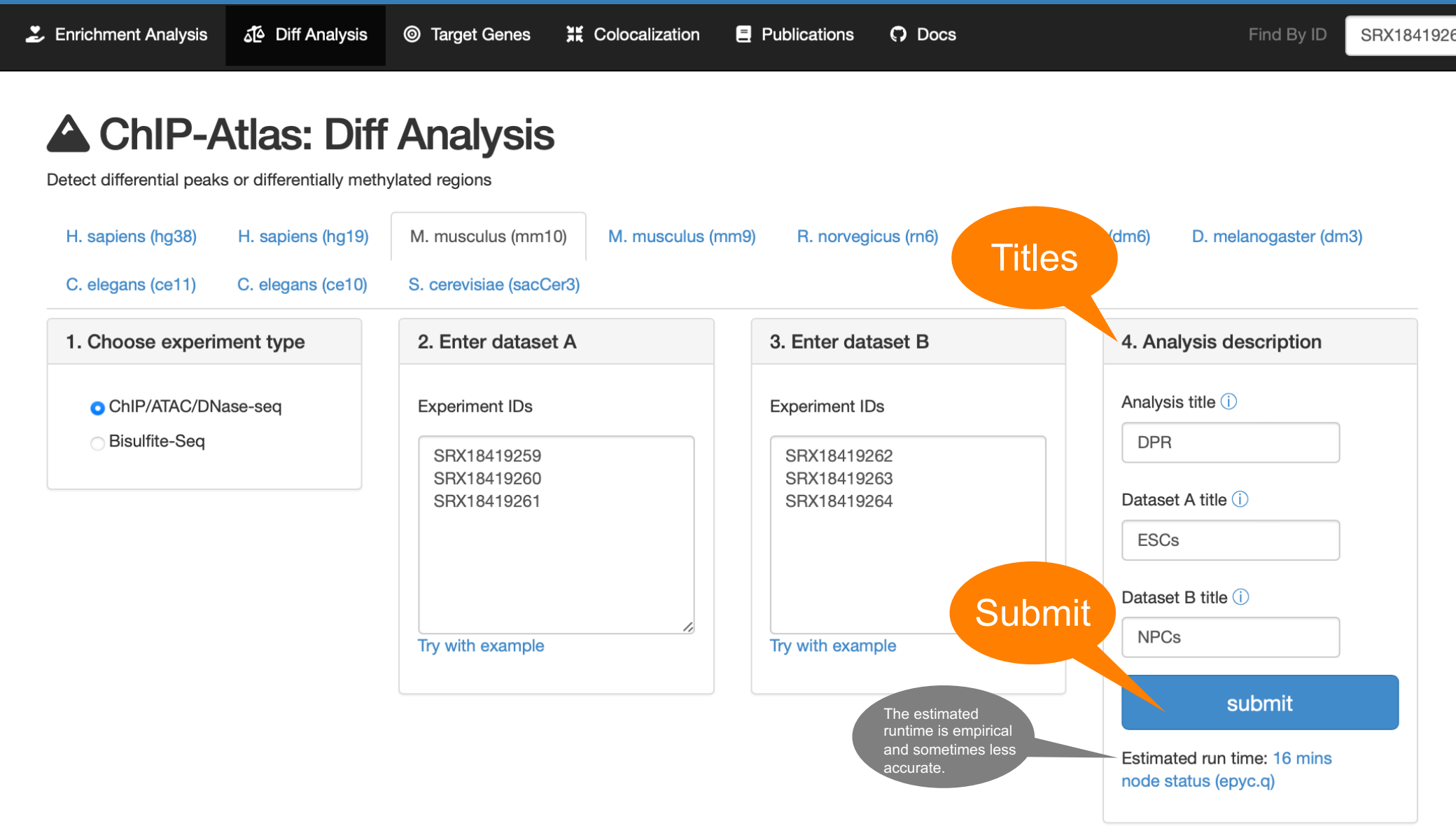
Settings to identify differential peak regions (DPRs)



Enter one or more IDs of your interest. For example, IDs of ATAC-seq data in ESCs (dataset A) and NPCs (dataset B) are entered.

Tips: IDs of experiments

Settings to identify differential peak regions (DPRs)



Enter arbitrary titles of this analysis (DPR), dataset A (ESCs), and dataset B (NPCs), which will be used to label the result data. Click on the “submit” button.

Running

Browser | Enrichment Analysis | Diff Analysis | Target Genes | Colocalization | Publications | Docs | Find By ID | GSM469863

ChIP-Atlas: Diff Analysis

Detect differential peaks or differentially methylated regions

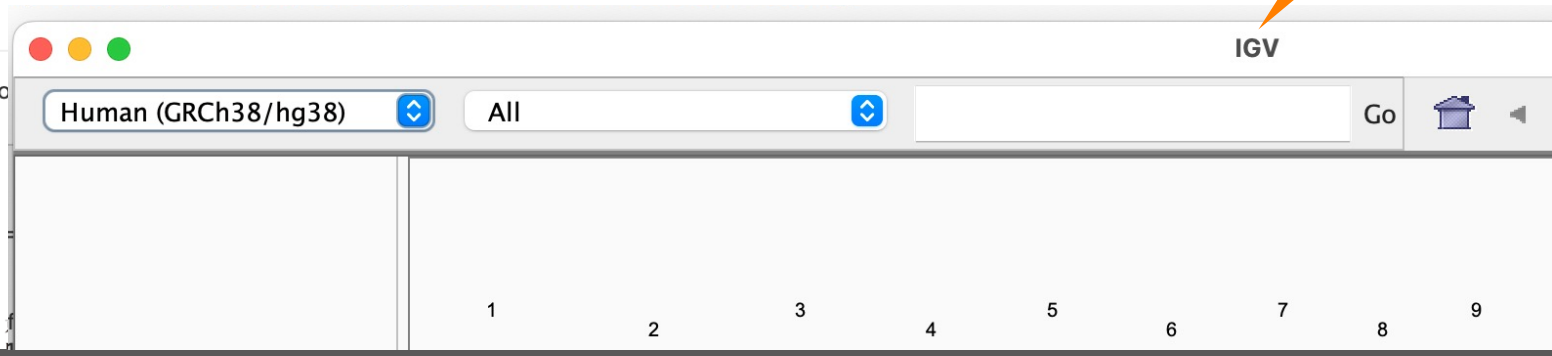
Result page URL will be available for a week from the time when 'status' is 'finished'.

Project title	DPR
Request ID	wabi_chipatlas_2024-0110-1436-50-311-488822
Submitted at:	14:36:50 (Jan-10-2024)
Estimated finishing time:	14:52:50 (Jan-10-2024)
Current time:	14:41:31 (Jan-10-2024)
Status	running
View on IGV:	http://localhost:60151/load?file=https://chip-atlas.dbcls.jp/data/query/wabi_chipatlas_2024-0110-1436-50-311-488822.igv.bed&genon...
Download Result:	https://chip-atlas.dbcls.jp/data/query/wabi_chipatlas_2024-0110-1436-50-311-488822.zip

running

Launch IGV

As this function uses the NIG superco



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Launch the IGV Genome Browser and wait several minutes while the “Status” is “running”.

Tips: Installation of IGV

Viewing the results

Browser | Enrichment Analysis | Diff Analysis | Target Genes | Colocalization | Publications | Docs | Find By ID | GSM469863

ChIP-Atlas: Diff Analysis

Detect differential peaks or differentially methylated regions

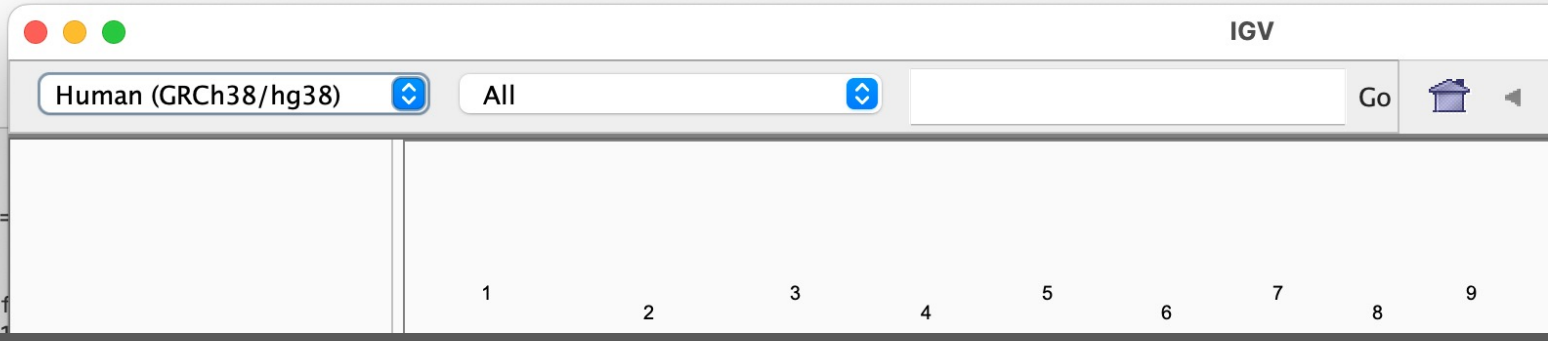
Result page URL will be available for a week from the time when 'status' is 'finished'.

Project title	DPR
Request ID	wabi_chipatlas_2024-0110-1436-50-311-488822
Submitted at:	15:17:15 (Jan-10-2024)
Estimated finishing time:	15:33:15 (Jan-10-2024)
Current time:	15:18:37 (Jan-10-2024)
Status	finished
View on IGV:	http://localhost:60151/load?file=https://chip-atlas.dbcls.jp/data/query/wabi_chipatlas_2024-0110-1436-50-311-488822.igv.bed&genome=mm10
Download Result:	https://chip-atlas.dbcls.jp/data/query/wabi_chipatlas_2024-0110-1436-50-311-488822.zip

finished

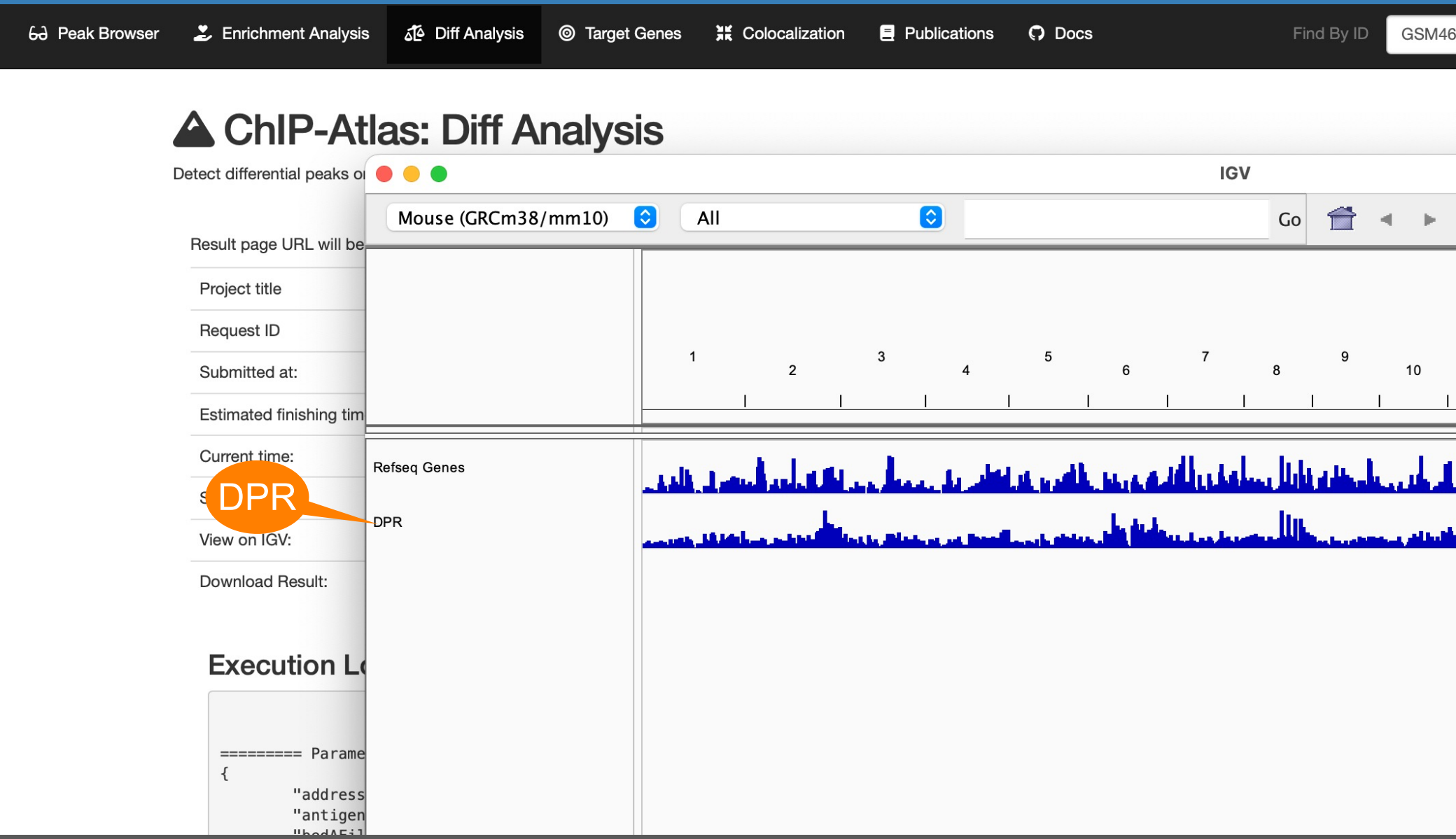
Execution Log

```
========= Parameters =======
{
        "address": null,
        "antigenClass": "dif
```



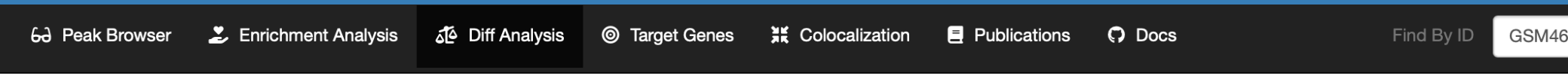
If you only want to browse the DPRs, click on the URL displayed next to “View on IGV”.

Viewing the results



DPRs are loaded into the IGV window.

Viewing the results



ChIP-Atlas: Diff Analysis

Detect differential peaks or differentially methylated regions

Result page URL will be available for a week from the time when 'status' is 'finished'.

Project title	DPR
Request ID	wabi_chipatlas_2024-0110-1436-50-311-488822
Submitted at:	15:17:15 (Jan-10-2024)
Estimated finishing time:	15:33:15 (Jan-10-2024)
Current time:	15:35:02 (Jan-10-2024)
Status	finished
View on IGV:	http://localhost:60151/load?file=https://chip-atlas.dbcls.jp/data/query/wabi_chipatlas_2024-0110-1436-50-311-488822.igv.bed&genome=mm10
Download Result:	https://chip-atlas.dbcls.jp/data/query/wabi_chipatlas_2024-0110-1436-50-311-488822.zip

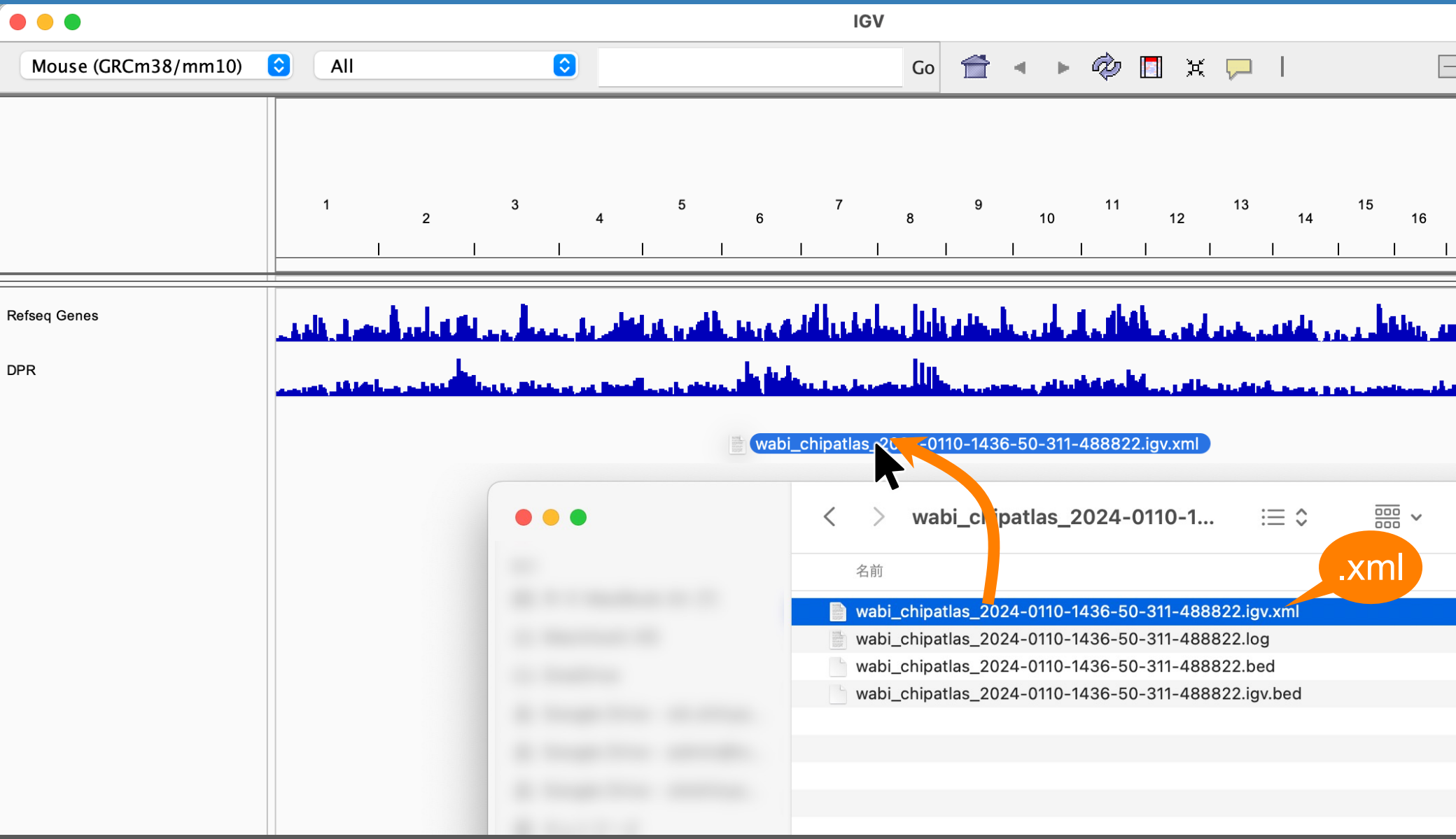
Execution Log

```
========= Parameters ========
{
        "address": null,
        "antigenClass": "diffbind",
```

If you want to browse the DPRs along with the original alignment data, click on the URL displayed next to “Download Result”.

Link to the download

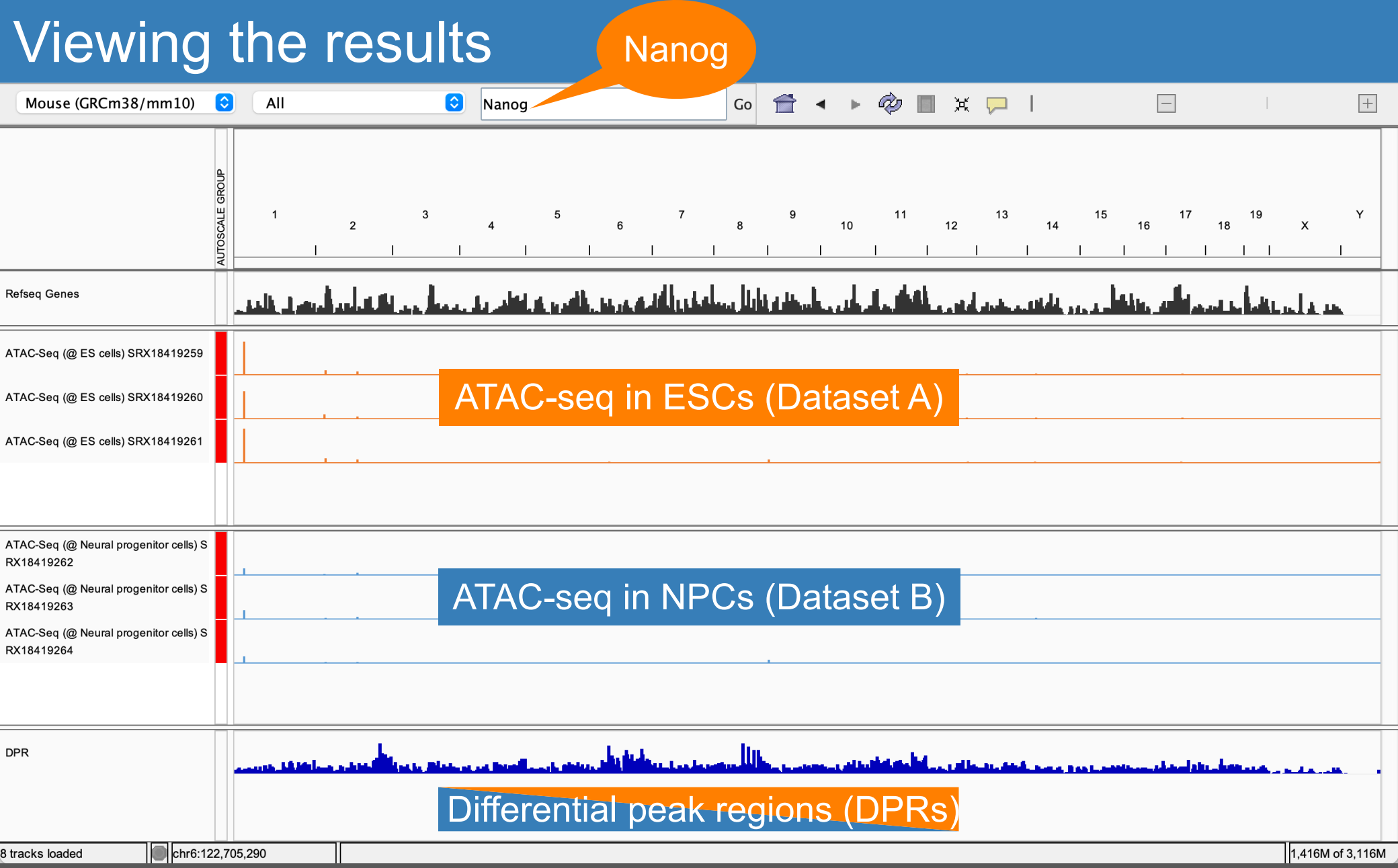
Viewing the results



Unzip the downloaded ZIP file, then drag and drop the IGV session file (.xml) onto the IGV window.

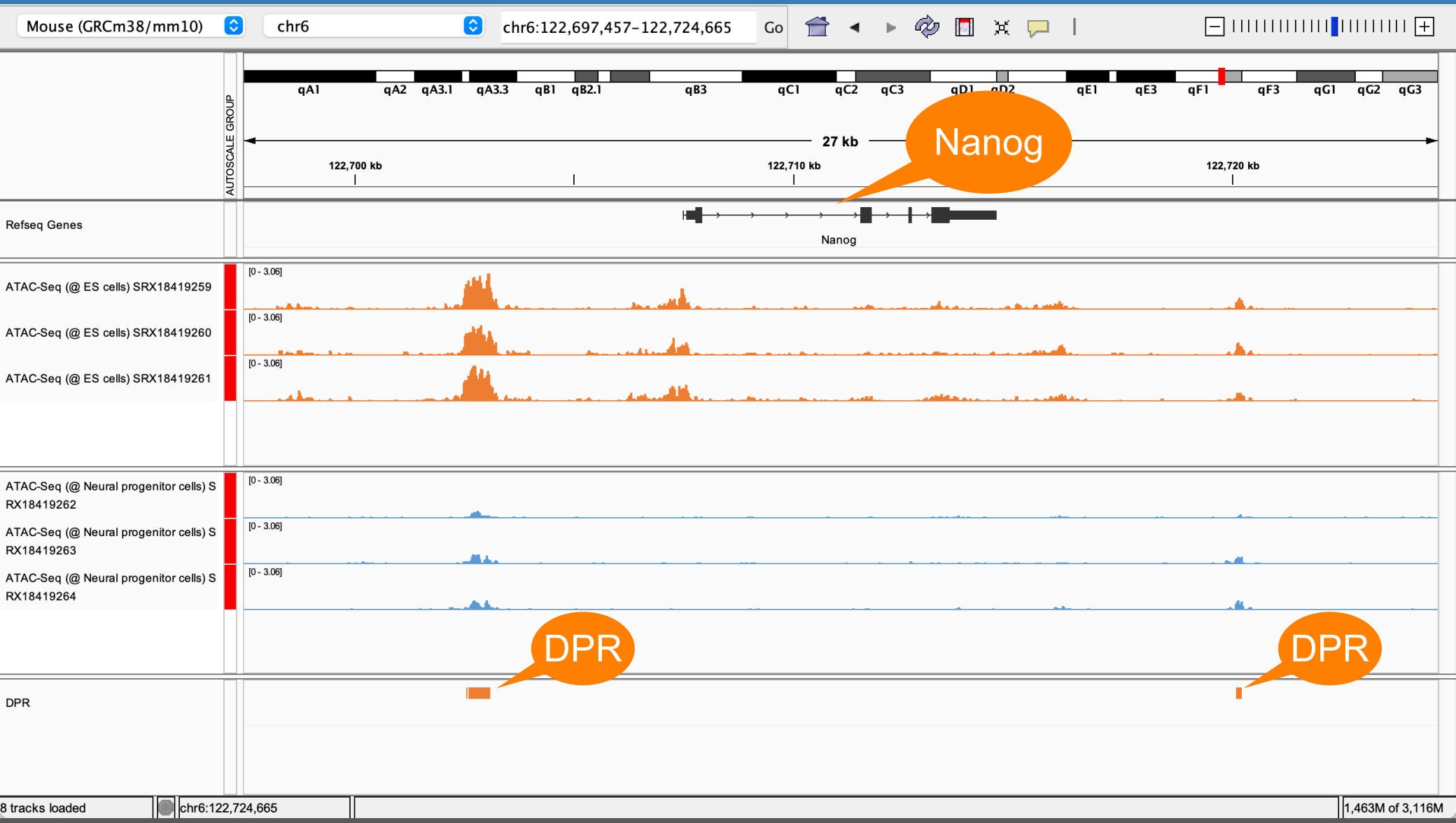
Tips: Downloaded files

Viewing the results



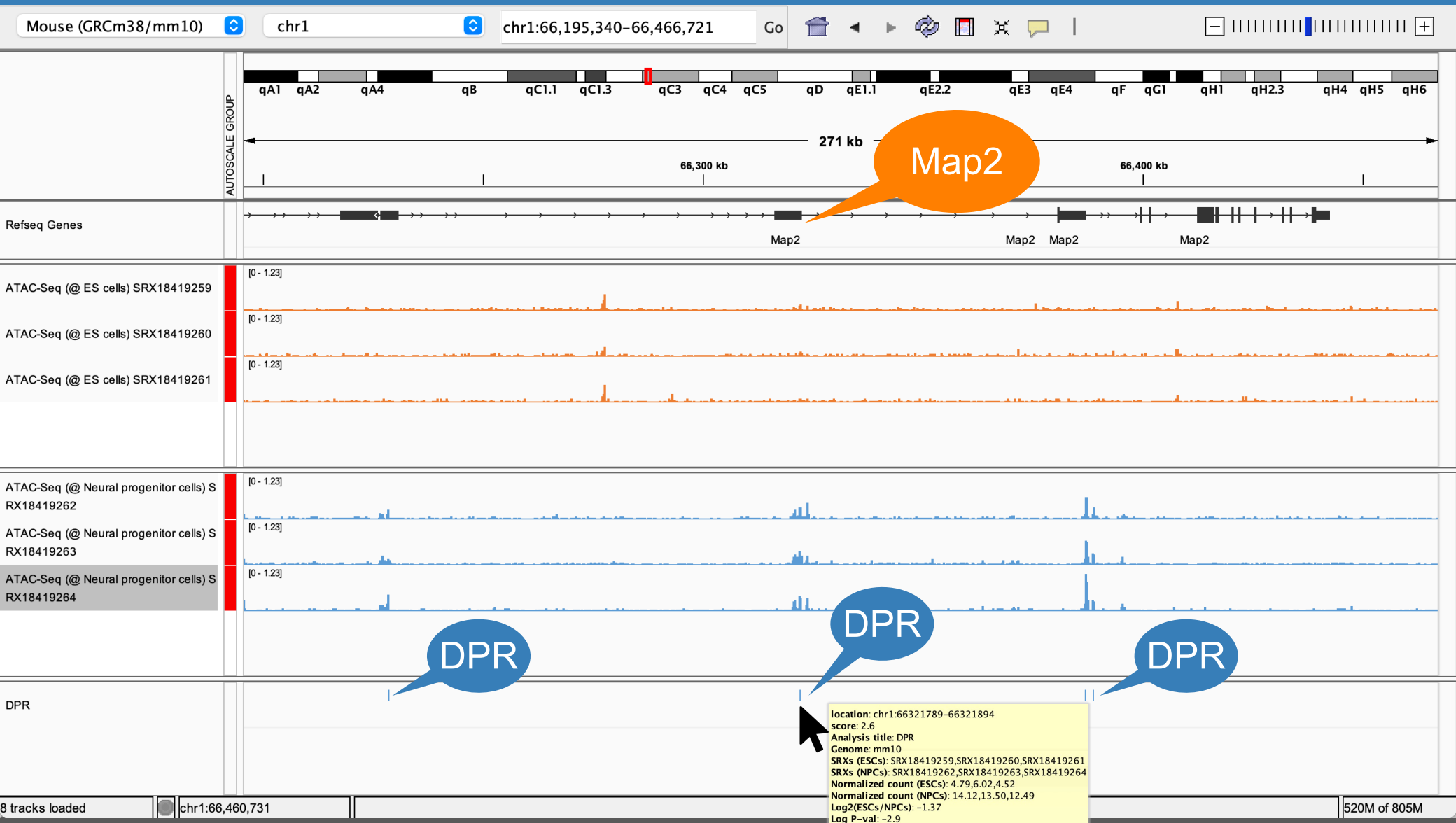
Now, you can browse DPRs along with the original data for comparison. Enter the gene of your interest (e.g., *Nanog*, a gene specifically expressed in ES cells).

Viewing the results



You can see the ESC-specific peak regions around the *Nanog* locus.

Viewing the results



You can see the NPC-specific peak regions around the *Map2* locus. Detailed information is displayed when the mouse pointer is hovered over the peaks.

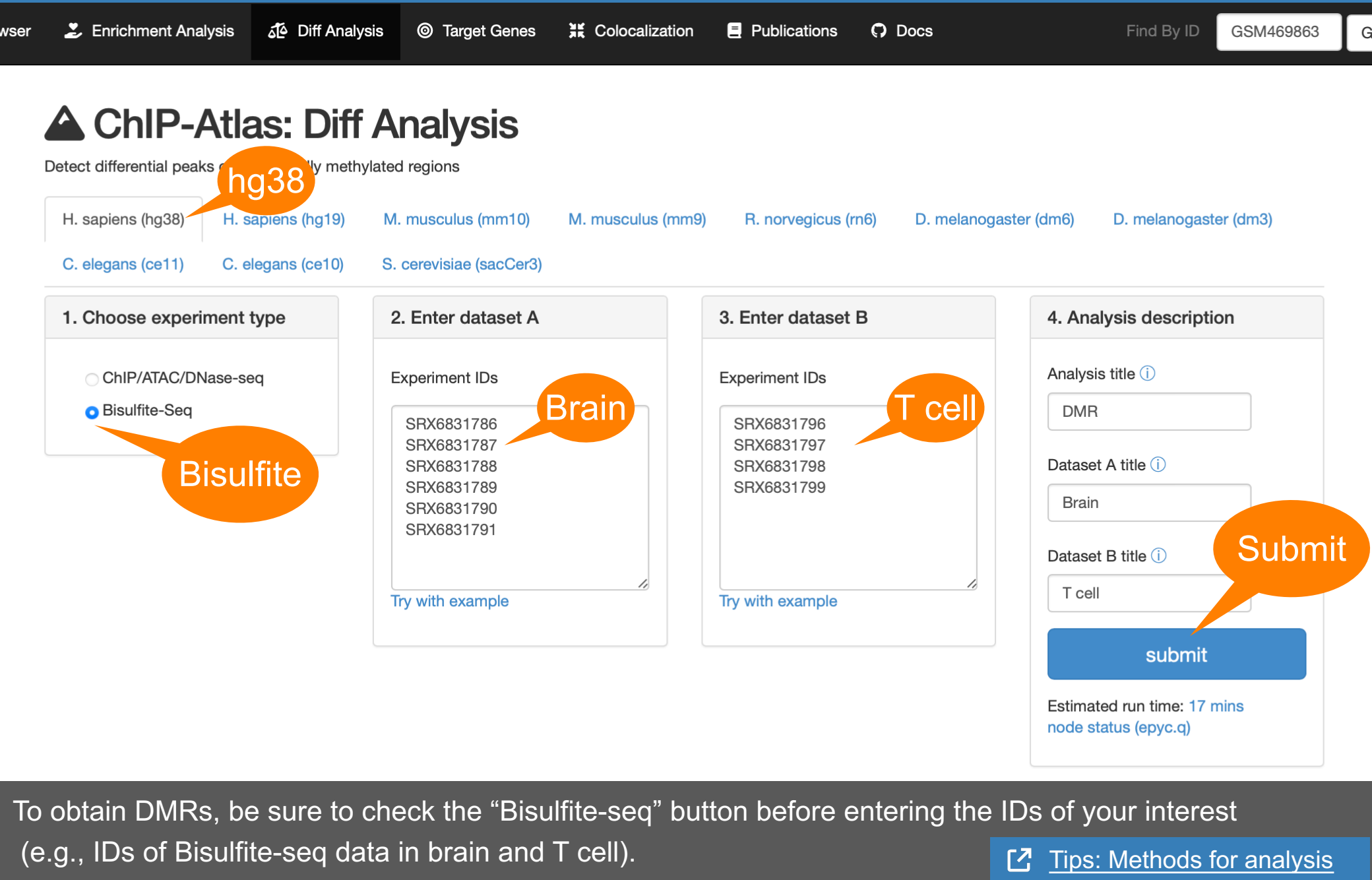
The manual for

ChIP-Atlas: Diff Analysis

1. Overview
2. Differential peak regions
   (ChIP-seq/ATAC/DNase-seq data)
3. Differentially methylated regions
   (Bisulfite-seq data)
4. Tips

Next, let us try to identify differentially methylated regions (DMRs) from two sets of Bisulfite-seq data in human brains and T cells.

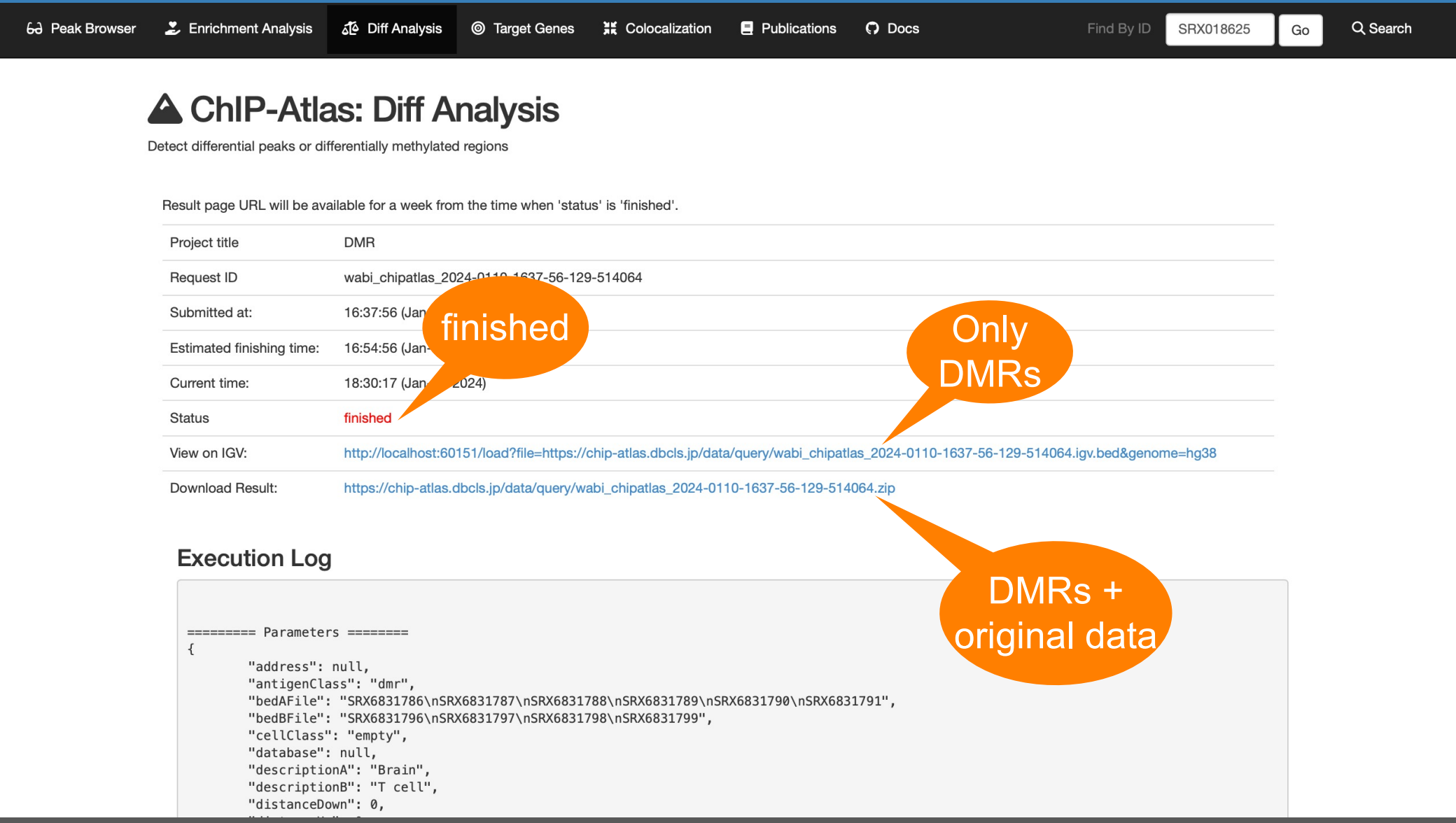
Settings to identify differentially methylated regions (DMRs)



To obtain DMRs, be sure to check the “Bisulfite-seq” button before entering the IDs of your interest (e.g., IDs of Bisulfite-seq data in brain and T cell).

Tips: Methods for analysis

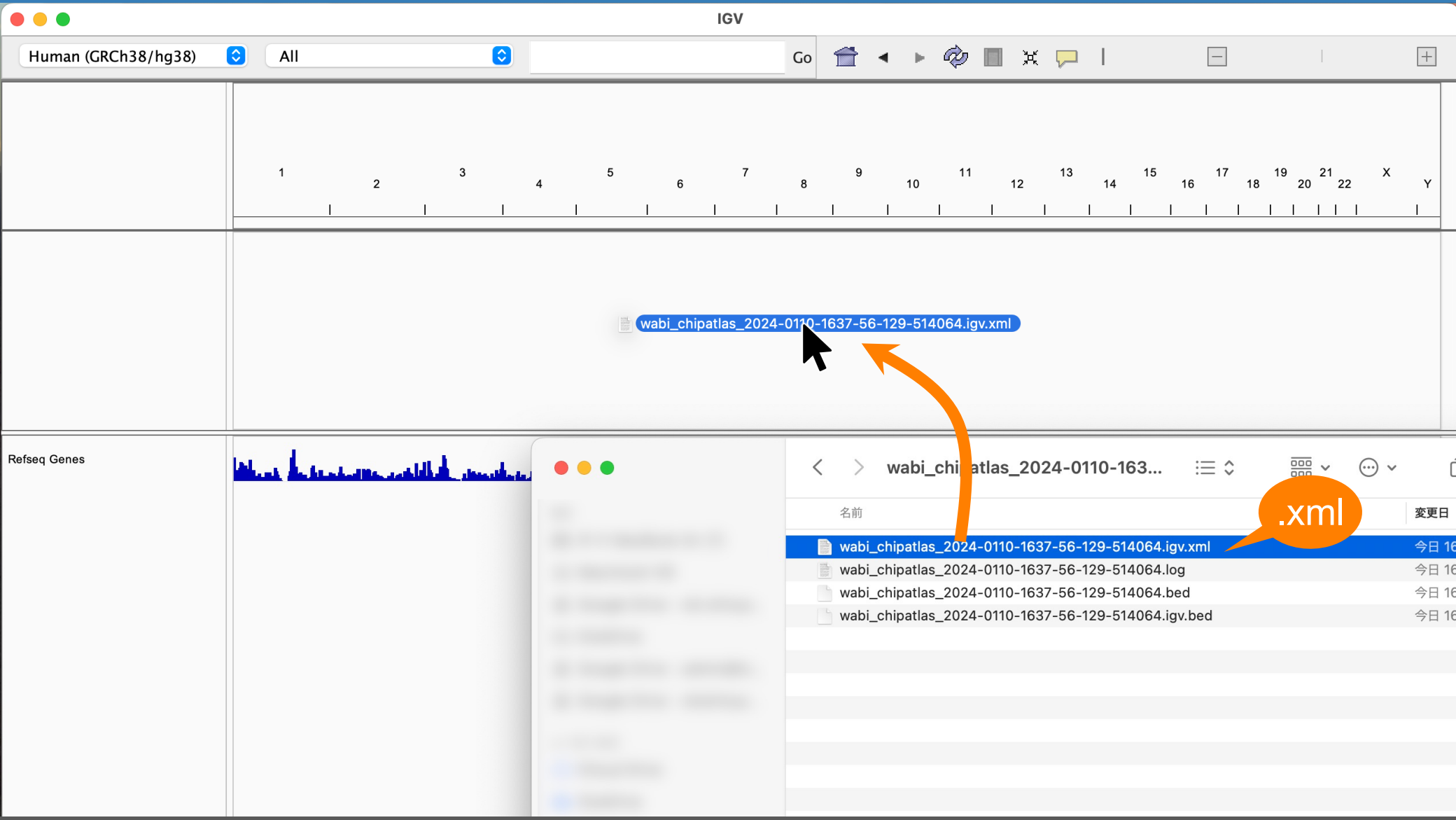
Settings to identify differentially methylated regions (DMRs)



When the analysis has finished, you can browse DMRs by clicking the URL displayed next to “View in IGV” and original Bisulfite-seq data by clicking the URL next to “Download Result”.

Link to the download

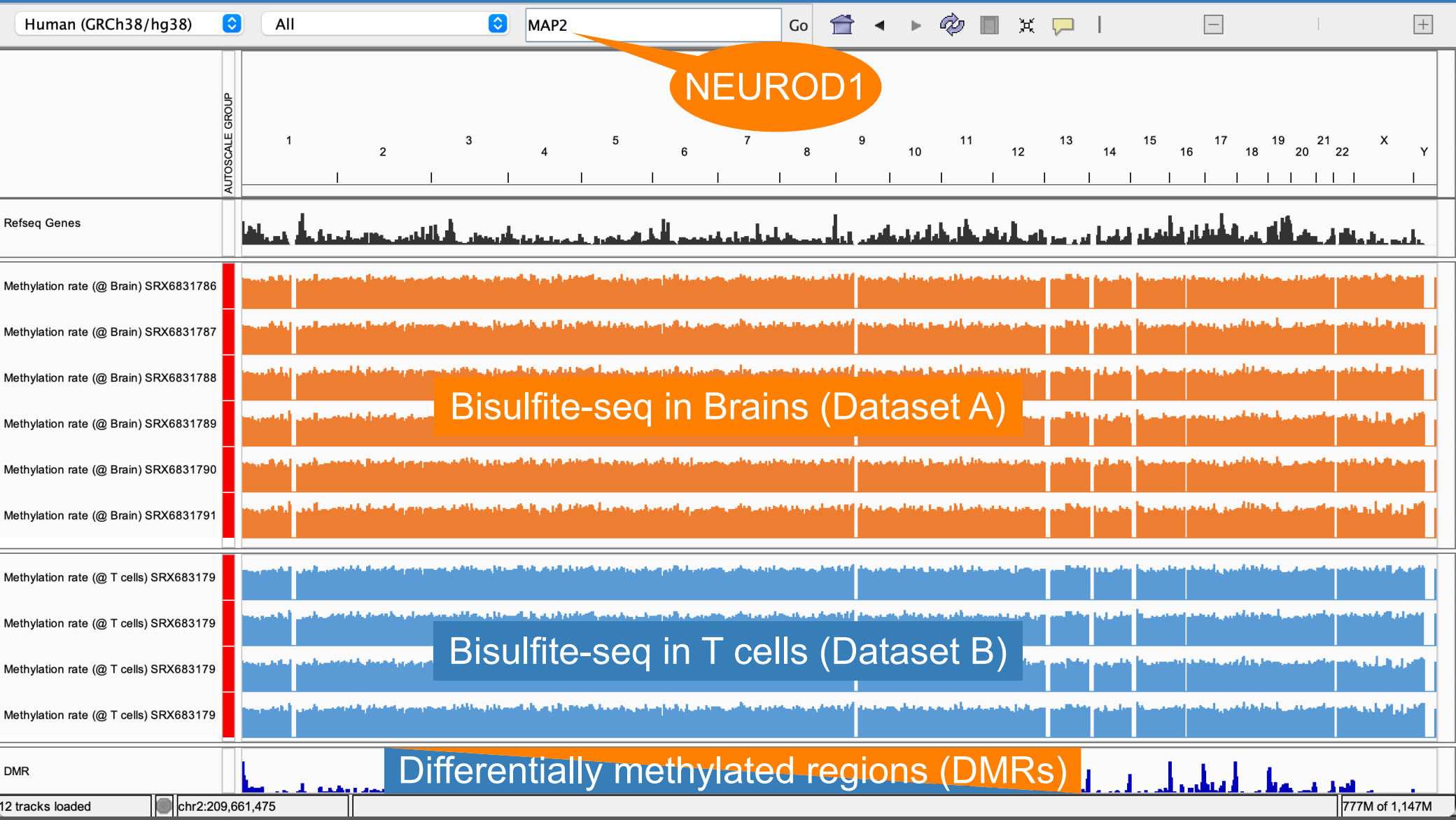
Settings to identify differentially methylated regions (DMRs)



Unzip the downloaded zip file, then drag and drop the IGV session file (.xml) onto the IGV window.

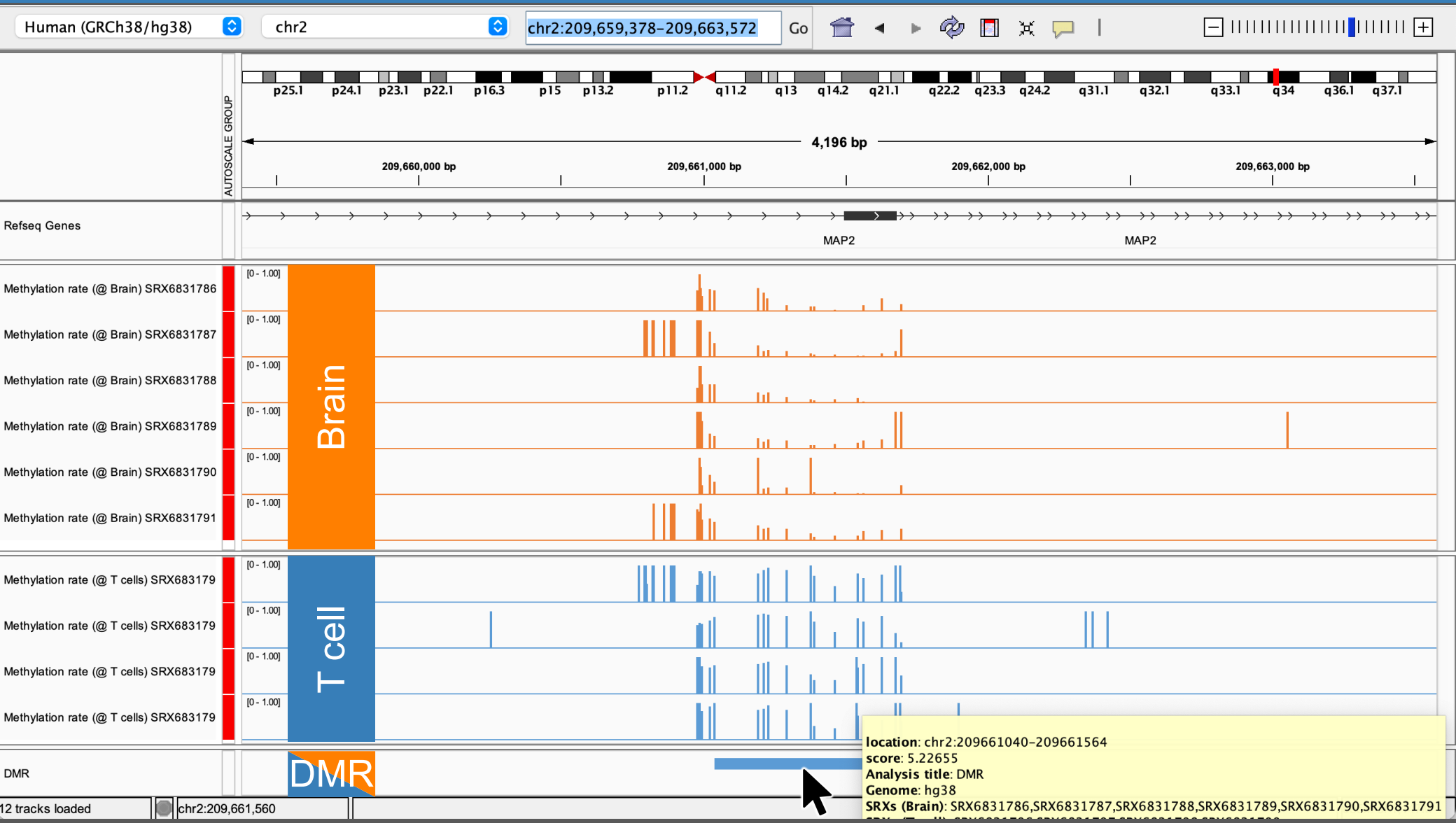
Tips: Downloaded files

Settings to identify differentially methylated regions (DMRs)



Now, you can browse the DMRs along with the original Bisulfite-seq data for comparison. Enter the gene of your interest (e.g., *MAP2*, a gene specifically expressed in neuronal cells).

Settings to identify differentially methylated regions (DMRs)



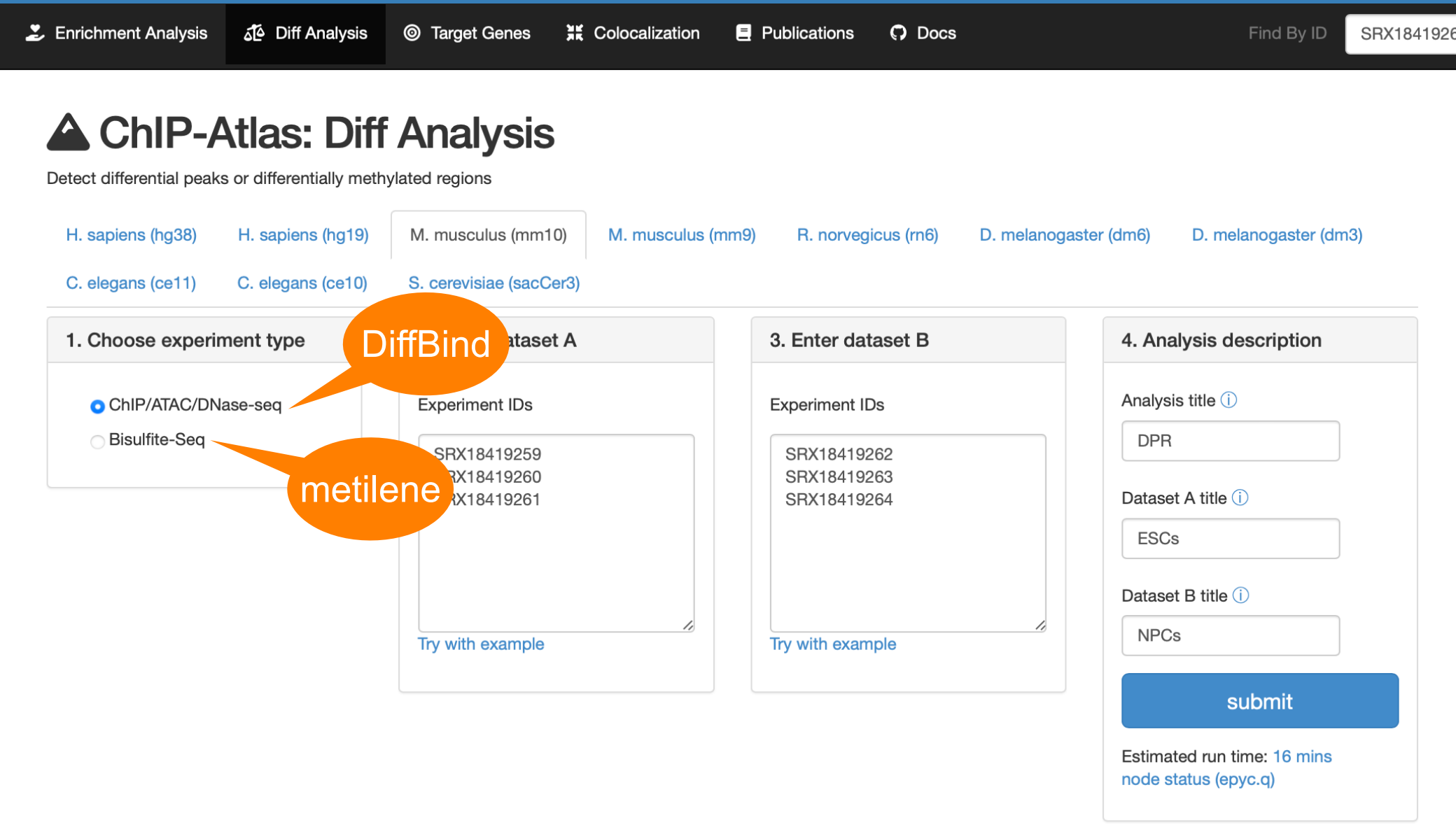
T cell-specific highly methylated regions are shown around the *MAP2* locus.

The manual for

ChIP-Atlas: Diff Analysis

1. Overview
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   (Bisulfite-seq data)
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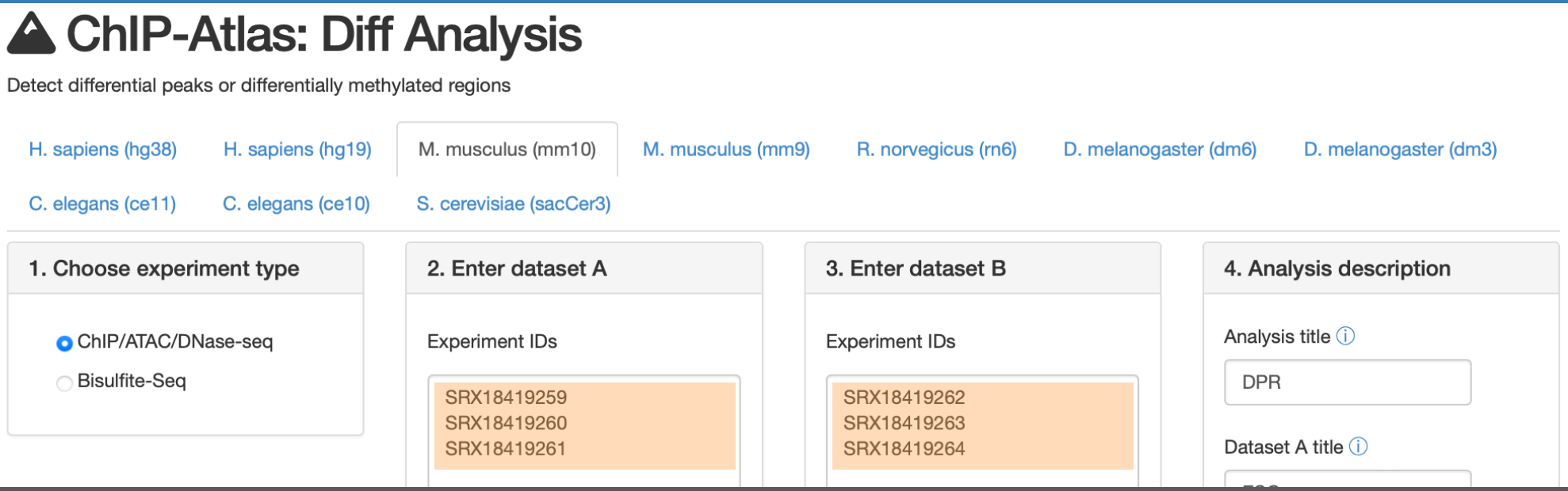
Tips: Methods for analysis



DPRs: Inspired by DiffBind (Stark, R. and Brown, G. 2011. bioconductor.org)

DMRs: metilene (Juhling, F., et al. 2016. Genome Res)

Tips: IDs of experiments



Acceptable data

- ID(s) of NCBI, EBI, and DDBJ (e.g., SRX18419259, ERX1103210, DRX335588)
- ID(s) of NCBI GEO (e.g., GSM6765200)
- Publicly accessible URL(s) of data on custom web server
  - ChIP/ATAC/DNase-seq: The URLs of raw read coverage (UCSC bigWig) in **integer** values, peak-call (UCSC BED) data, and total number of mapped reads. Try with the following example in ***hg38***.

```
_PREFIX_/sample_A1.bw<tab>_PREFIX_/sample_A1.bed<tab>205201674    # Dataset A
_PREFIX_/sample_A2.bw<tab>_PREFIX_/sample_A2.bed<tab>208332830
```

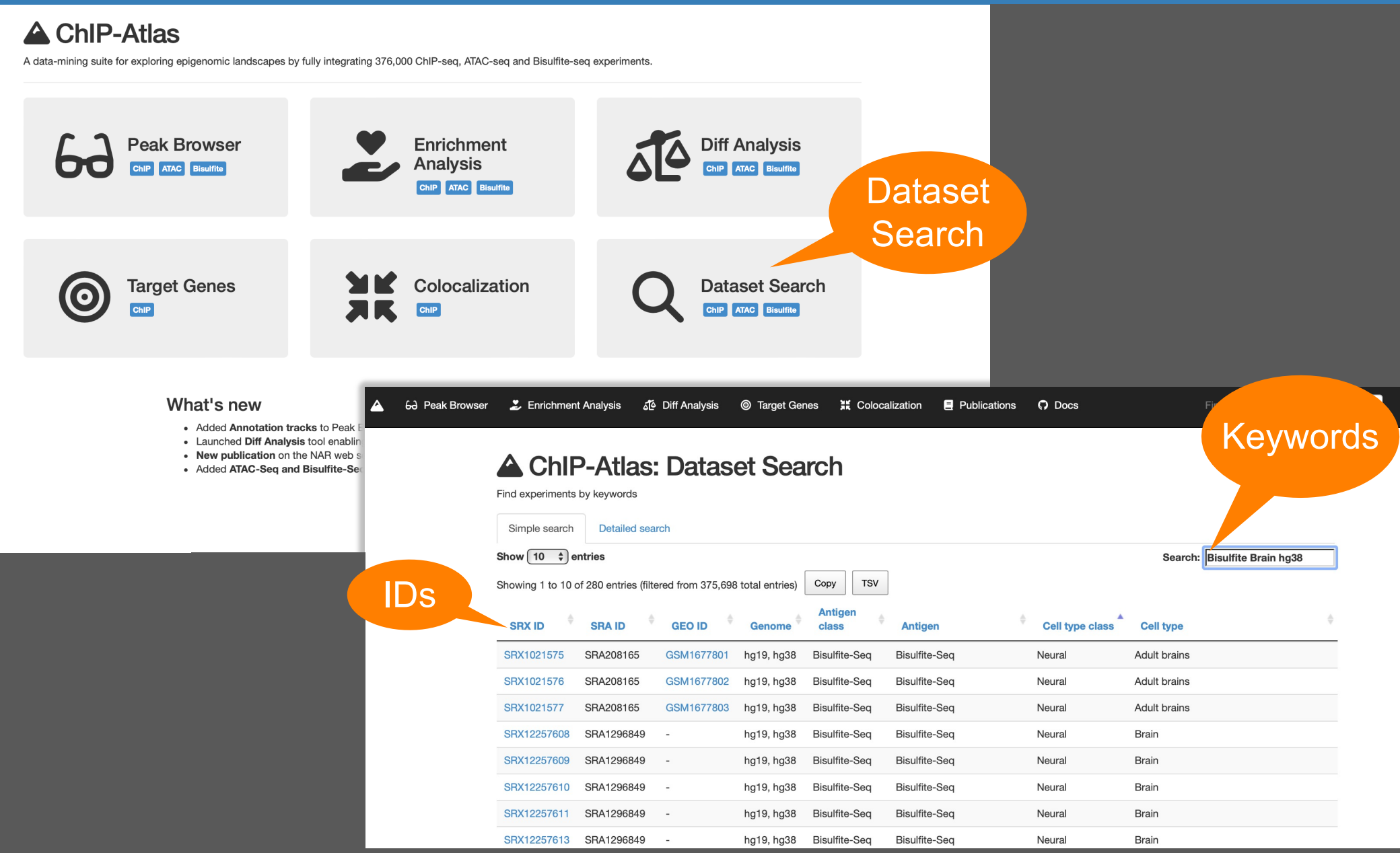
```
_PREFIX_/sample_B1.bw<tab>_PREFIX_/sample_B1.bed<tab>87753996    # Dataset B
_PREFIX_/sample_B2.bw<tab>_PREFIX_/sample_B2.bed<tab>90562744
```

  - Bisulfite-seq: bigWig data of methylation rate (between 0 and 1) . Try with the following example in ***rn6***.

```
_PREFIX_/sample_A3.bw    # Dataset A
_PREFIX_/sample_B3.bw    # Dataset B
```

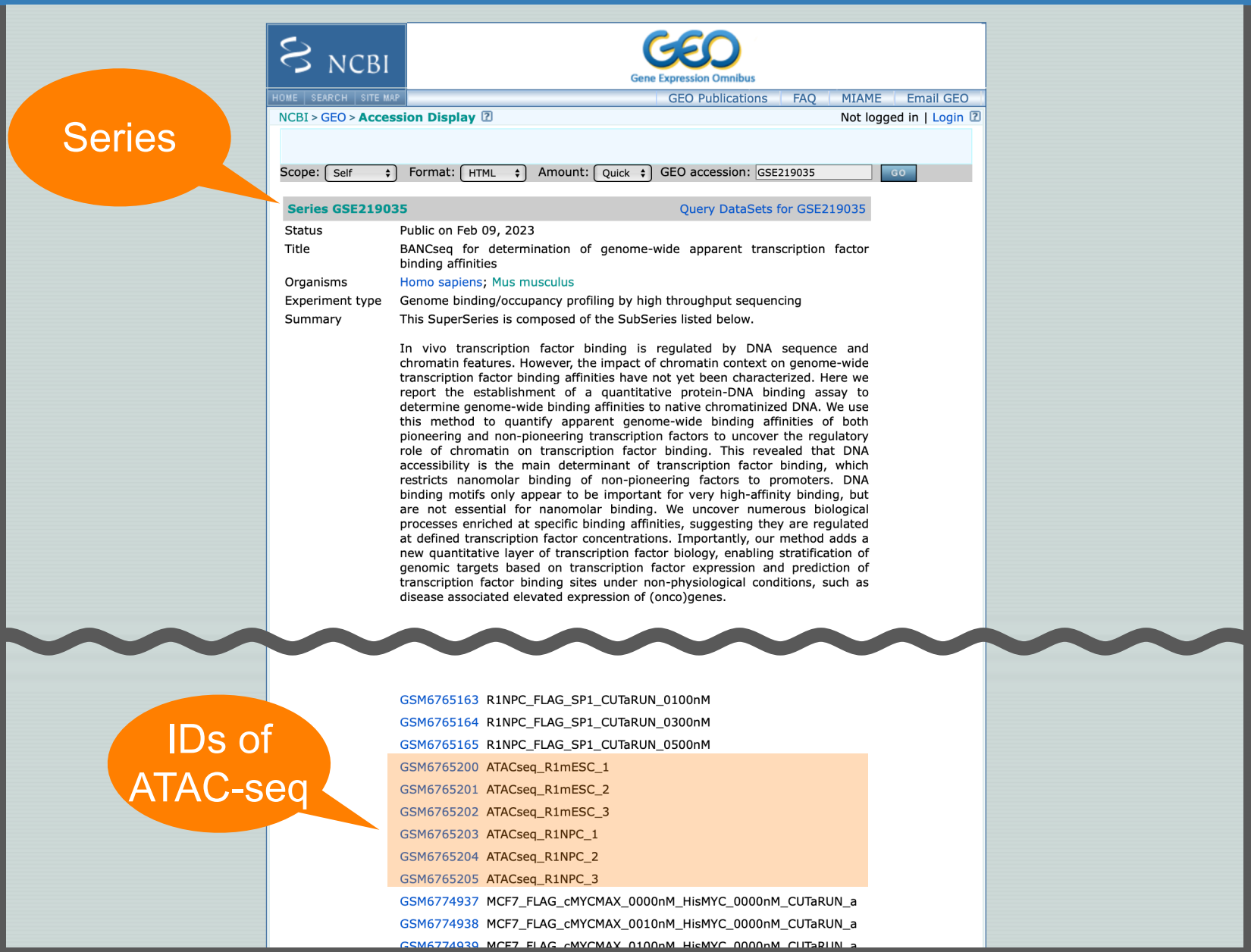
* Replace `_PREFIX_` with `https://chip-atlas.dbcls.jp/data/manual/examples`

Tips: IDs of experiments



The ChIP-Atlas Dataset Search tool is useful to find the IDs of your interest by keyword search.

Tips: IDs of experiments



NCBI GEO is also useful for finding the IDs of interest within an identical series of experiments.

Tips: Installation of IGV

Download IGV

Did you know that there is also an **IGV web application** that runs only in a web browser, does not use Java, and requires no downloads? See **https://igv.org/app**. Click on the Help link in the app for more information about using IGV-Web.

Latest release of IGV Desktop: 2.17.0

M1/M2 Mac Users:

Apple's Rosetta software is required to run the IGV MacOS App that includes Java. If you run IGV with your own Java installation, Rosetta may not be required if your version of Java runs natively on M1/M2.

Linux users:

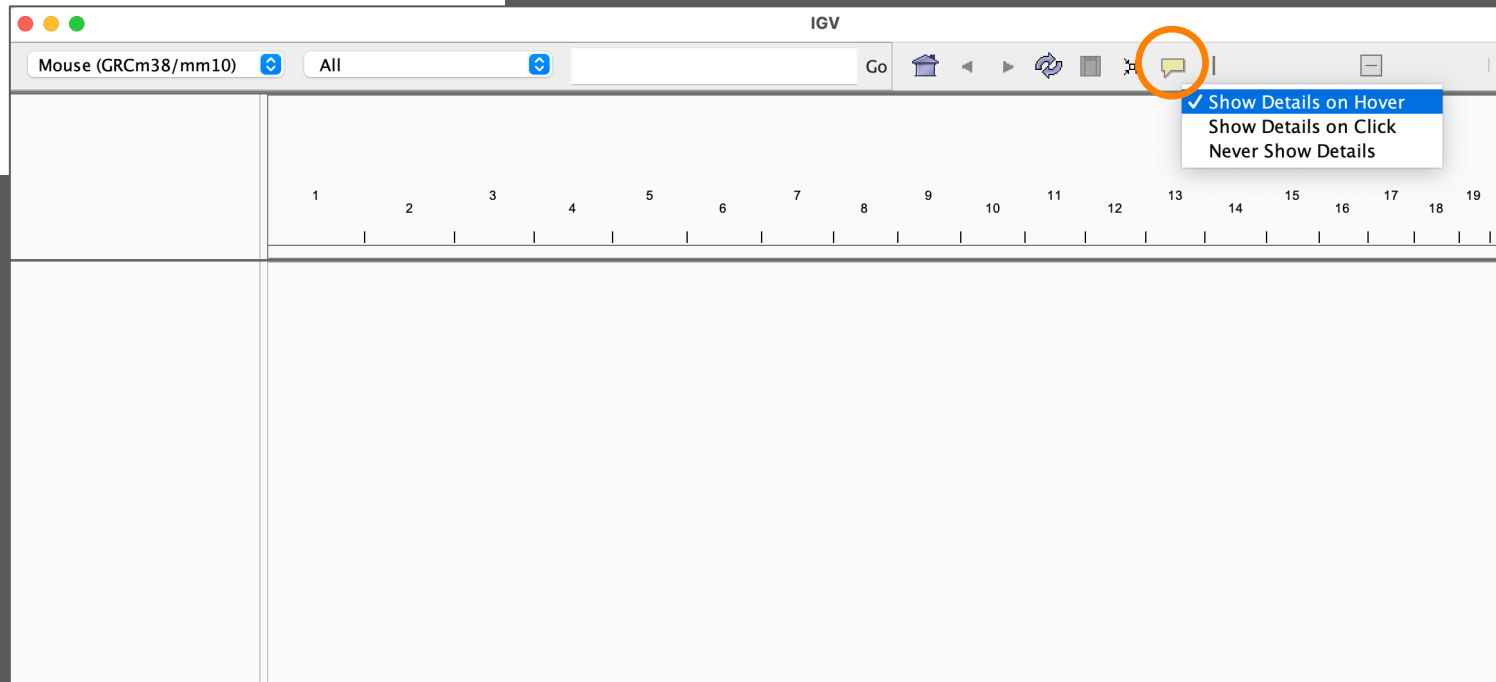
The 'IGV for Linux' download includes AdoptOpenJDK (now Eclipse Temurin) version 17 for x64 Linux. See their list of supported platforms here. If your platform is not on the "x64 Linux" list, or the packaged Java does not work on your version of Linux, download the 'Command line IGV for all platforms' and use it with your own Java installation.

What's New: See the Release Notes for what's new in each IGV release.

- IGV MacOS App — Java included
- IGV MacOS App — Separate Java 17 required
- IGV for Windows — Java included
- IGV for Windows — Separate Java 17 required
- IGV for Linux — Java included
- Command line IGV and igvtools for all platforms — Separate Java 17 required

← Download IGV (https://igv.org/doc/desktop/#DownloadPage/)

↓ Launch IGV and check “Show Details on Hover”



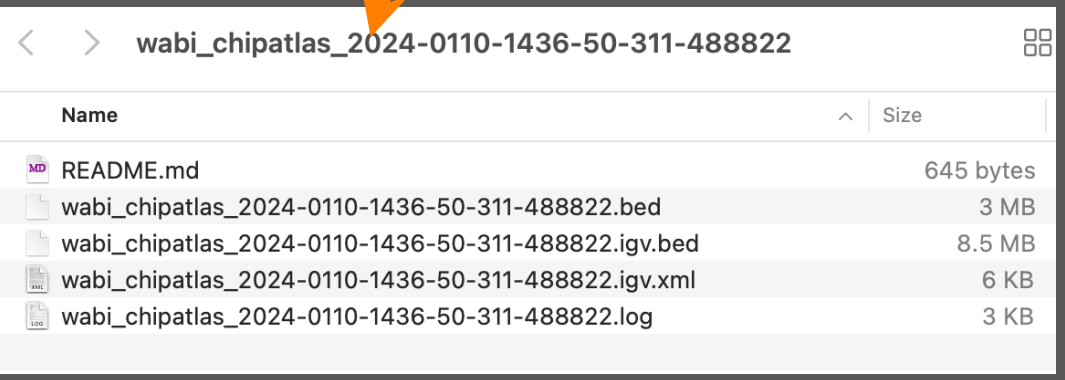
Tips: Downloaded files

ChIP-Atlas: Diff Analysis

Detect differential peaks or differentially methylated regions

Result page URL will be available for a week from the time when 'status' is 'finished'.

Project title	DMR
Request ID	wabi_chipatlas_2024-0110-1637-56-129-514064
Submitted at:	16:37:56 (Jan-10-2024)
Estimated finishing time:	16:54:56 (Jan-10-2024)
Current time:	18:30:17 (Jan-10-2024)
Status	finished
View on IGV:	http://localhost:60151/load?file=https://chip-atlas.dbcls.jp/data/query/wabi_chipatlas_2024-0110-1637-56-129-514064.igv.bed&genome=hg38
Download Result:	https://chip-atlas.dbcls.jp/data/query/wabi_chipatlas_2024-0110-1637-56-129-514064.zip



↑ Take a note of the URL so that you can review the results for up to one month.

README.md: Description of the downloaded files.

AnalysisID.igv.xml: IGV session file to display DPRs or DMRs together with the original data to be compared.

AnalysisID.log: Analysis log to identify DPRs or DMRs.

AnalysisID.bed: DPRs or DMRs in BED format suitable for further analysis.

AnalysisID.igv.bed: DPRs or DMRs in BED format suitable for browsing on IGV.